

ArcVault Intake & Triage

Working workflow: exported n8n workflow and screenshots of each step with output

Valsoft AI Engineer assessment, deliverable 4.1 n8n Cloud + Groq openai/gpt-oss-120b workflow version 1.0.1

This document shows the workflow running and what each step produced for the five sample requests in the brief.

Section A was captured in n8n Cloud from the live, published workflow. **Section B** lays out the output of every step for each sample; those pages are rendered directly from `records.json`, the records the live workflow returned, so no value in them was typed by hand. Sample #5 appears in both sections as the same record (`1cd7d42b...`), which ties the two together.

The pipeline, in order. Webhook → Normalize → **LLM** Classify → validate → **LLM** Enrich → validate → Route & Escalate → **LLM** Summarize → Assemble record → POST to queue → respond. The three LLM calls classify, extract and summarise. The routing and escalation decisions are made by deterministic code, so they are repeatable and can be explained without reference to a model.

- Of the five samples, **2 escalated** to the human review queue and 3 went straight to their owning team.
- Every escalated record keeps the queue it *would* have gone to, so the reviewer knows who to hand it to.

A. The workflow running in n8n Cloud

These five screenshots are the evidence that the exported JSON is what actually ran.

Figure 1. The published workflow (12 nodes)

The canvas, left to right, as listed above. The branch below the main line is the **Has Text?** check: an empty message skips both LLM calls and goes straight to routing, where it is escalated rather than dropped. The **Published** badge at the top right means the production webhook URL is live.

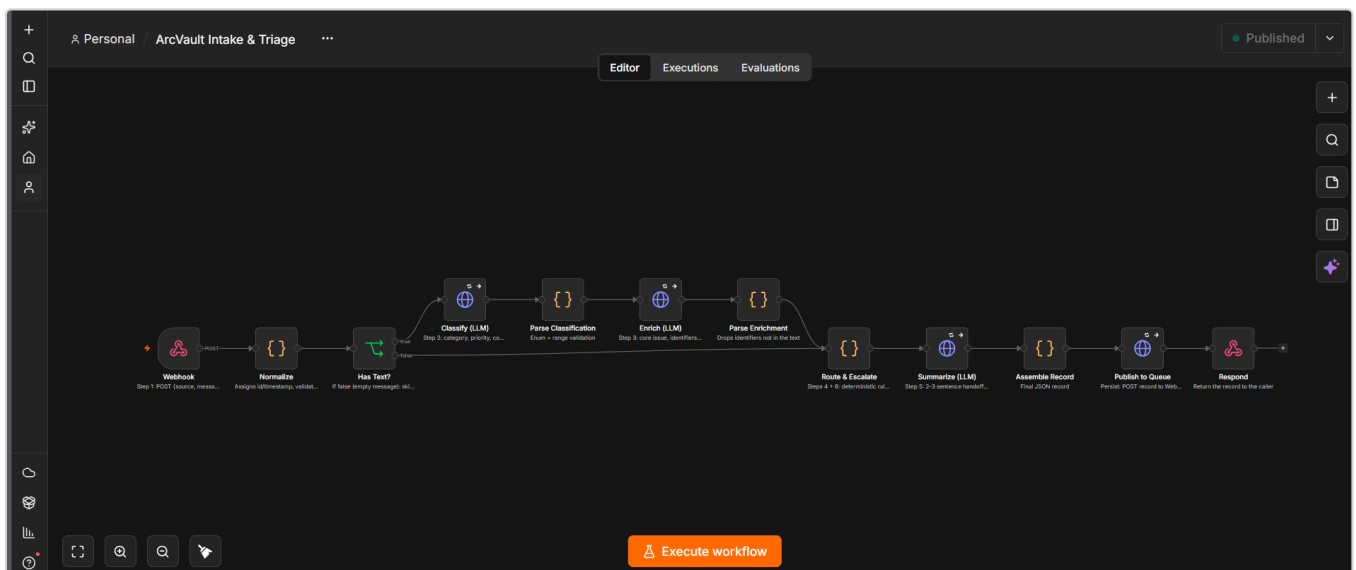


Figure 2. Execution history

One row per inbound message, each about four seconds end to end, with every node green. Opening a run shows the input and output of every node, which is where the workflow's state lives for debugging.

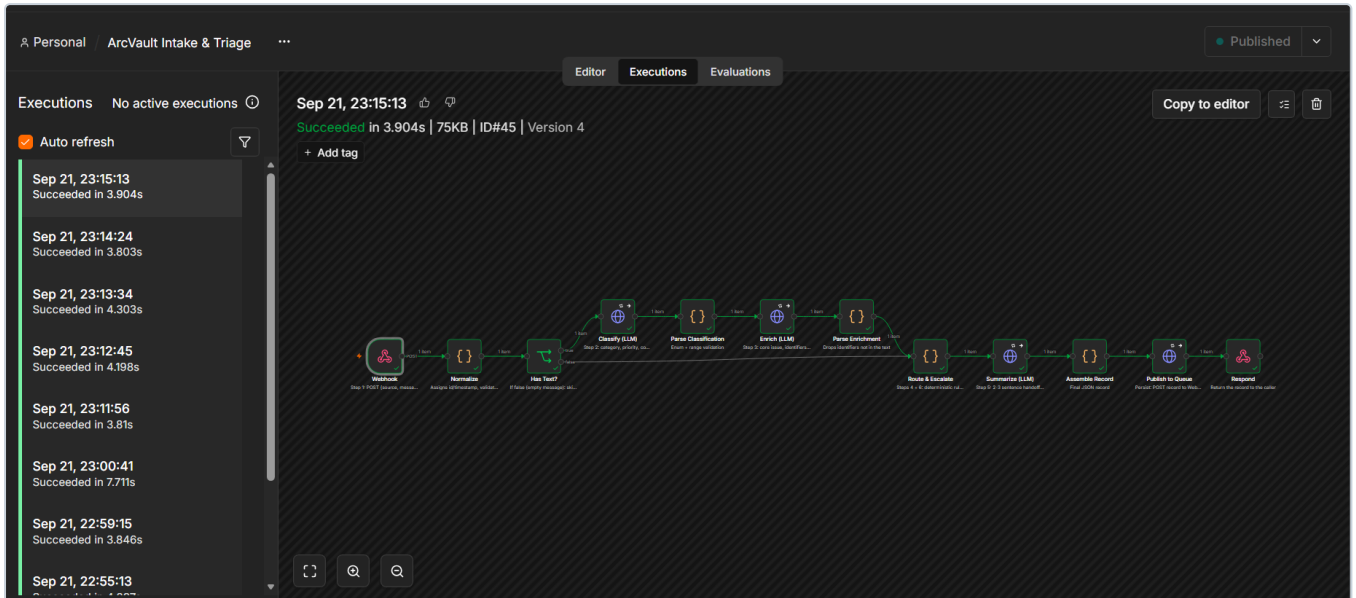


Figure 3. Sample #3: the escalation decision (Route & Escalate node)

The output of the deterministic routing step for the invoice dispute. `standard_queue` is `billing`, but `destination_queue` is `human_review_escalation` because `billing_amount_over_threshold` fired. The `detail` line records exactly why: the largest amount found in the text was \$1,240 against a \$500 threshold. No LLM is involved in this decision.

```
"decision": {
  "priority": "Medium",
  "priority_adjusted_by": null,
  "routing": {
    "standard_queue": "billing",
    "destination_queue": "human_review_escalation",
    "routing_reason": "Escalated for human review (billing_amount_over_threshold);
                      standard owner would be billing."
  },
  "escalation": {
    "flagged": true,
    "triggered_rules": [
      {
        "rule": "billing_amount_over_threshold",
        "detail": "basis=any_amount; largest amount $1240, discrepancy $260,
                  threshold $500"
      }
    ],
    "advisory_signals": [
    ]
  }
},
}
```

Figure 4. Sample #5: the final record, part 1 (Assemble Record node)

The first half of the record for the outage report: the raw message, the classification with its confidence and rationale, and the extracted identifiers. Identifiers that do not appear literally in the message are dropped, and `dropped_identifiers` would list them.

```
[
  {
    "record_id": "1cd7d42b-bf76-4d15-9d74-5ed7136941d8",
    "received_at": "2026-09-22T16:51:33.902Z",
    "source": "web_form",
    "raw_message": "Your dashboard stopped loading for us around 2pm EST. Checked our end – it's definitely on yours. Multiple users affected.",
    "classification": {
      "category": "Incident/Outage",
      "secondary_categories": [
      ],
      "priority": "High",
      "llm_priority": "High",
      "priority_adjusted_by": null,
      "confidence": 0.96,
      "rationale": "Multiple users affected and dashboard stopped loading for us"
    },
    "enrichment": {
      "core_issue": "The customer reports that the dashboard stopped loading and multiple users are affected.",
      "identifiers": {
        "account_ids": [
        ],
        "invoice_numbers": [
        ],
        "error_codes": [
        ],
        "urls": [
        ],
        "monetary_amounts": [
        ],
        "products_or_features": [
          "dashboard"
        ],
        "other": [
          "2pm EST"
        ]
      },
      "dropped_identifiers": [
      ],
      "urgency_signal": {
        "level": "High",
        "evidence": [
          "stopped loading",
          "Multiple users affected"
        ]
      }
    }
  },
]
```

Figure 5. Sample #5: the final record, part 2

The second half: routing, the two escalation rules that fired with their reasons, the summary written for the receiving team, and the processing metadata (model, prompt versions, latency, workflow version). This is the complete payload POSTed to the queue and returned to the caller.

```
[{"routing": {"standard_queue": "incident_response", "destination_queue": "human_review_escalation", "routing_reason": "Escalated for human review (incident_category, outage_language); standard owner would be incident_response."}, "escalation": {"flagged": true, "triggered_rules": [{"rule": "incident_category", "detail": "category is Incident/Outage"}, {"rule": "outage_language", "detail": "matched: stopped loading, multiple users affected"}], "advisory_signals": []}, "summary": "The customer reports that the dashboard stopped loading around 2pm EST and multiple users are affected. This is a high-priority incident affecting several users. Escalate to human review for immediate investigation of the dashboard outage.", "processing": {"model": "openai/gpt-oss-120b", "prompt_versions": {"classification": "classification.v2", "enrichment": "enrichment.v2", "summary": "summary.v1"}, "latency_ms": 2020, "workflow_version": "1.0.1", "llm_errors": []}}]
```

B. Step-by-step output for the five sample requests

One page per sample. Each page shows the six workflow steps in the brief's order, labelled with the n8n node that produced each one, so classification, enrichment, routing, escalation and the final summary can be read together. The table below is the same five records side by side.

Figure 6. All five records at a glance

Category, priority, confidence, extracted identifiers, the standard and destination queue, the escalation rules that fired, and the summary sent to the receiving team. Rows 3 and 5 are the two escalations; the arrow shows the queue the ticket would have gone to without one.

ArcVault triage — structured output for the 5 sample requests

Live n8n run · rendered from output/records.json · model openai/gpt-oss-120b

#	Category	Priority	Conf.	Identifiers	Standard → destination queue	Escalated (rules)	Summary
#1 email	Bug Report	High	0.94	arcvault.io/user/jsmith 403 last Tuesday	engineering ↓ engineering	no	The customer cannot log in and receives a 403 error on account arcvault.io/user/jsmith, which started after the update last Tuesday. This is a high-priority issue affecting access to the service. Next step: engineering should investigate the authentication failure linked to the recent update.
#2 web_form	Feature Request	Low	0.94	bulk export audit logs	product ↓ product	no	The customer requests a bulk export feature for audit logs. They note that, as a compliance-heavy organization, this would save them hours each month. Add this request to the product backlog for future prioritization.
#3 support_portal	Billing Issue	Medium	0.95	#8821 \$1,240 \$980/month	billing ↓ human_review_escalation	yes billing_amount_over_threshold	Invoice #8821 shows a charge of \$1,240 but the contract rate is \$980/month, a \$260 discrepancy. This billing difference exceeds the \$500 escalation threshold, so it requires immediate human review. Action: assign to a billing specialist to investigate and adjust the invoice as needed.
#4 email	Technical Question	Low	0.92	SSO Okta	technical_support ↓ technical_support	no	The customer asks whether SSO with Okta can be set up for their organization. This is a low-priority technical question about authentication options. Next step: provide the standard documentation and steps for configuring Okta SSO in ArcVault.
#5 web_form	Incident/Outage	High	0.96	dashboard 2pm EST	incident_response ↓ human_review_escalation	yes incident_category outage_language	The customer reports that the dashboard stopped loading around 2pm EST and multiple users are affected. This is a high-priority incident affecting several users. Escalate to human review for immediate investigation of the dashboard outage.

Figure 7. Sample #1 (email): every step

"Hi, I tried logging in this morning and keep getting a 403 error. My account is arcvault.io/user/jsmith. This started after your update last Tuesday."

A login failure for one named account. It tests the Bug Report vs Incident/Outage boundary (one user, not the service) and the security_signal rule: a 403 raises priority but does not escalate on its own, because escalating every login problem would flood the human queue. The advisory signal is still recorded.

Result: Bug Report, priority High, confidence 0.94; routed to engineering with no rule fired. Advisory: security_signal (recorded, not an escalation).

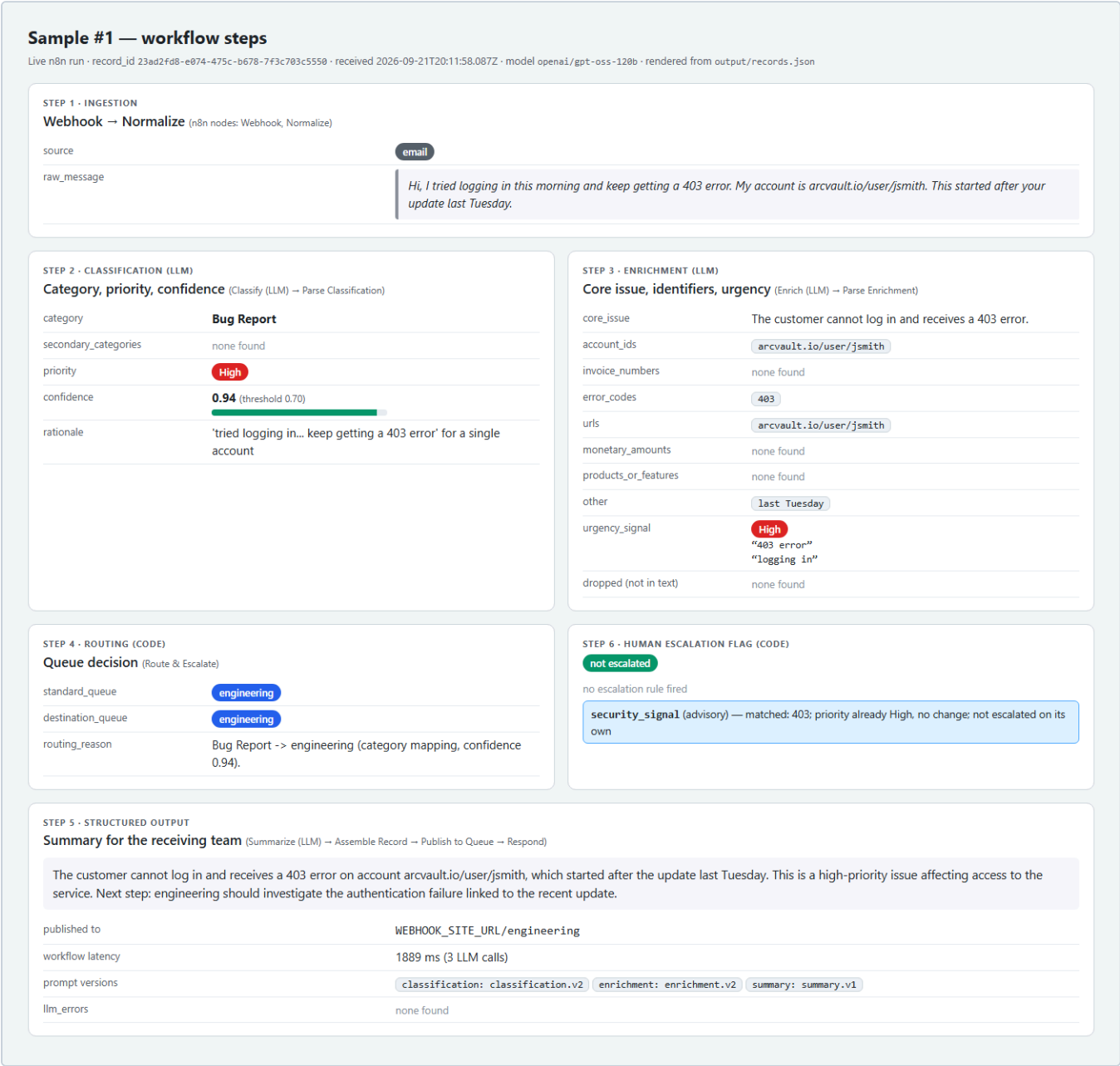


Figure 8. Sample #2 (web_form): every step

"We'd love to see a bulk export feature for our audit logs. We're a compliance-heavy org and this would save us hours every month."

A plain feature request, and the easiest case. It is here to show the normal path: no rule fires, so the record goes straight to the owning queue with no human in the loop.

Result: Feature Request, priority Low, confidence 0.94; routed to product with no rule fired.

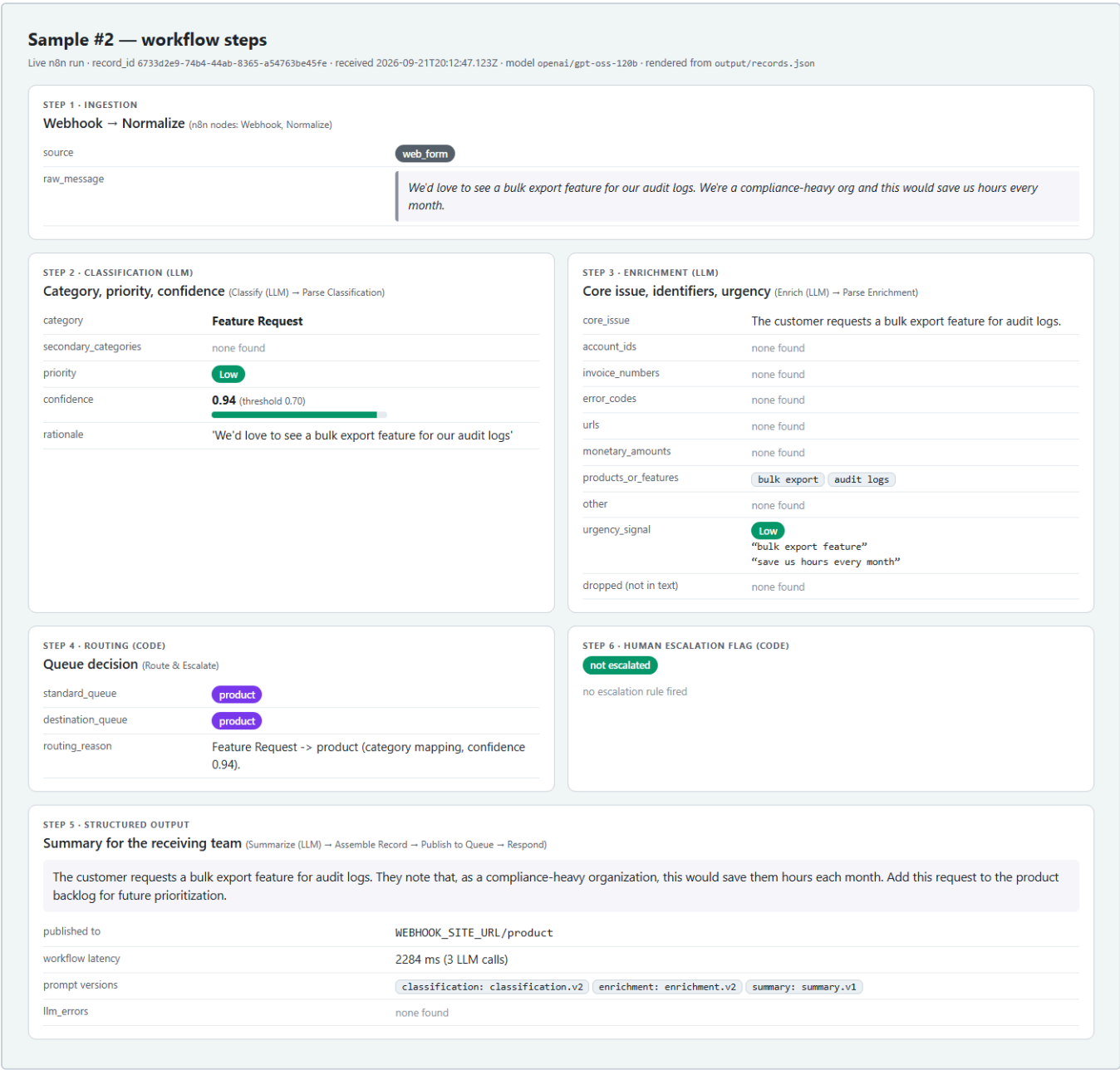


Figure 9. Sample #3 (support_portal): every step

"Invoice #8821 shows a charge of \$1,240 but our contract rate is \$980/month. Can someone look into this?"

An invoice dispute, and the main escalation case. The dollar amounts are parsed from the raw text by code, not taken from the LLM, so a hallucinated number cannot trigger or suppress the escalation.

Result: Billing Issue, priority Medium, confidence 0.95; escalated to `human_review_escalation` by `billing_amount_over_threshold`; the standard owner, `billing`, is kept in the record.

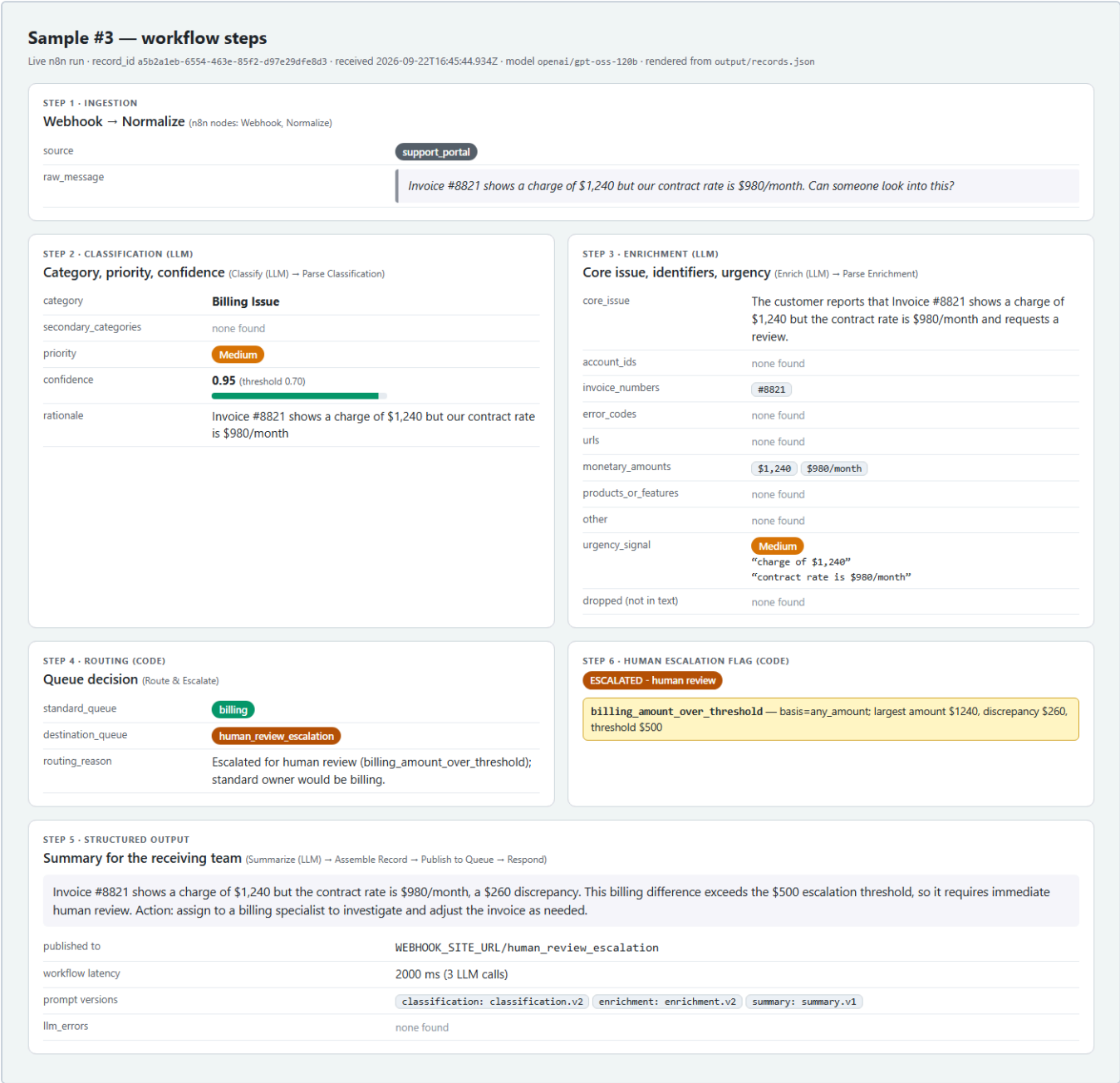


Figure 10. Sample #4 (email): every step

"I'm not sure if this is the right place to ask, but is there a way to set up SSO with Okta? We're evaluating switching our auth provider."

A pre-sales question about SSO. It tests the Technical Question vs Feature Request boundary ("is there a way" rather than "please build"), and confirms that merely mentioning SSO is not treated as a security signal.

Result: Technical Question, priority Low, confidence 0.92; routed to `technical_support` with no rule fired.

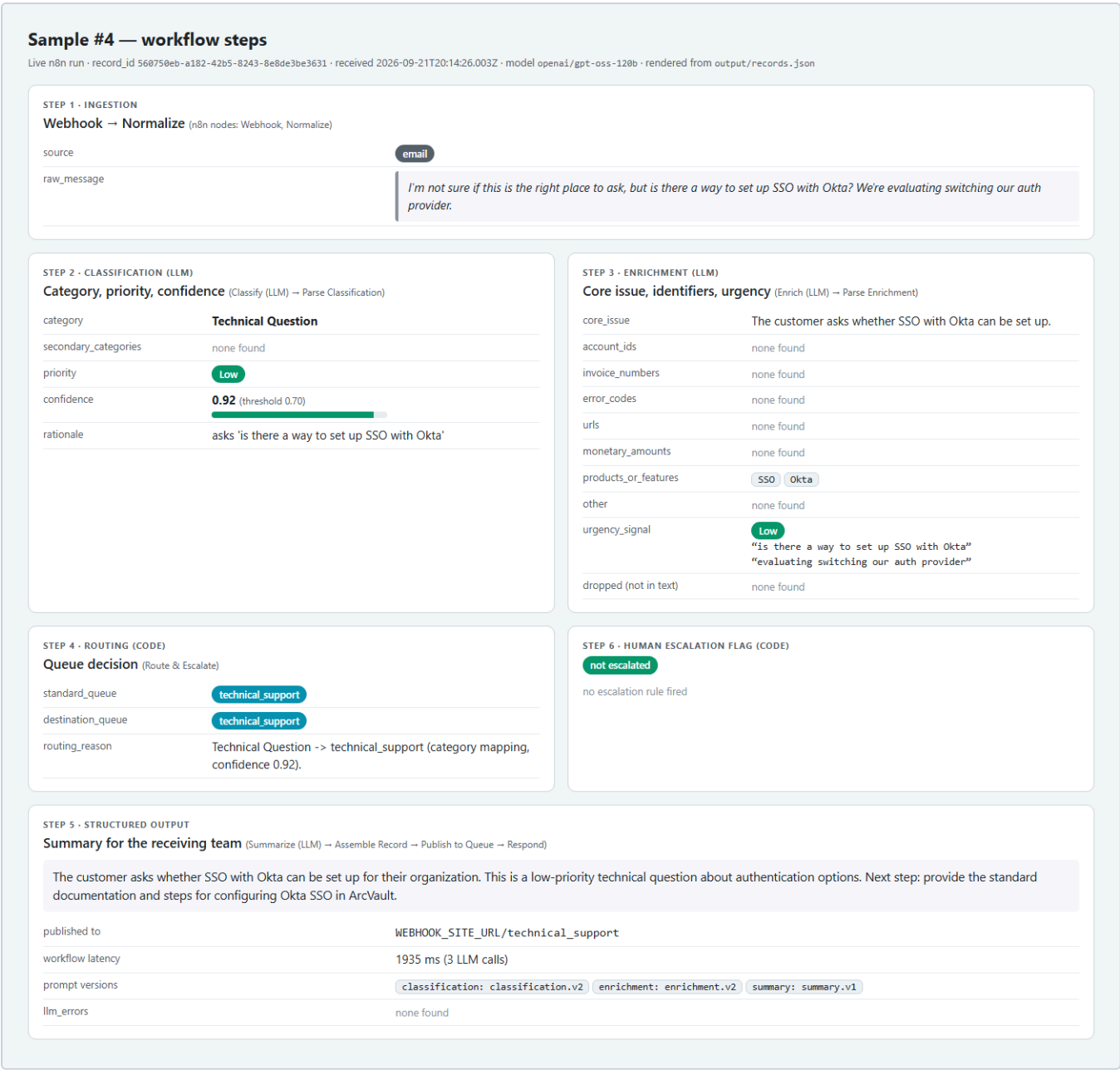


Figure 11. Sample #5 (web_form): every step

"Your dashboard stopped loading for us around 2pm EST. Checked our end — it's definitely on yours. Multiple users affected."

An outage report. Two independent rules fire: one from the category, one from phrase matching on the raw text. The second is a safety net that still escalates an outage even if the model had labelled it a Bug Report.

Result: Incident/Outage, priority High, confidence 0.96; escalated to `human_review_escalation` by `incident_category`, `outage_language`; the standard owner, `incident_response`, is kept in the record.

